# Ch. 15 HW #2 – Solutions

Principles of Microeconomics

December 8, 2025

1. Leah's variable cost per day is

$$\$280 - \$30 = \$250.$$

Therefore, Leah's average variable cost is

$$\$250/10 = \$25.$$

That's less than her average revenue of $27, so Leah produces in the short run.

In the long run, Leah's average total cost is

$$\$280/10 = \$28.$$

That's greater than her average revenue, $27, so Leah exits in the long run.

2. (a)

| Q | FC | VC | AFC | AVC | ATC | MC | Profit |
|---|---|---|---|---|---|---|---|
| 0 | 100 | 0 | – | – | – | – | -100 |
| 1 | 100 | 50 | 100 | 50 | 150 | 50 | -100 |
| 2 | 100 | 70 | 50 | 35 | 85 | 20 | -70 |
| 3 | 100 | 90 | 33.33 | 30 | 63.33 | 20 | -40 |
| 4 | 100 | 140 | 25 | 35 | 60 | 50 | -40 |
| 5 | 100 | 200 | 20 | 40 | 60 | 60 | -50 |
| 6 | 100 | 360 | 16.67 | 60 | 76.67 | 160 | -160 |

(b) See the profit column above. No, shutting down is not a wise decision because their losses would be greater from shutting down than they would be from producing 4 cases.

(c) The firm's losses are $100. It's not the best decision because if the firm produces 4 cases – the other quantity at which marginal cost equals marginal revenue – its losses are smaller, $40. That's because the firm's marginal costs decline at first, so the firm's marginal costs are actually less than its marginal revenue for cases 2 & 3.

3. In a competitive market, all of a firm's output gets sold at the market price. That means marginal revenue is equal to average revenue which is equal to the market price of the good, Therefore, the average revenue and the price of the good both equal \$10. Since the price is \$10, that means the firm sold \$500/\$10 = \$50

4. (a) Price equals average revenue which equals \$10. Therefore, profit is

$$\pi = (P - ATC) * Q = (10 - 8) * 100 = 200.$$

   (b) Since the firm is profit maximizing, marginal cost equals marginal revenue which equals the price of the good, \$10.

   (c) Average variable cost is

$$AVC = ATC - AFC = ATC - FC/Q = 8 - 200/100 = \$6.$$

   (d) Since marginal cost is greater than average total cost, average total cost is greater than its minimum. That means the firm's efficient scale is less than 100 units.